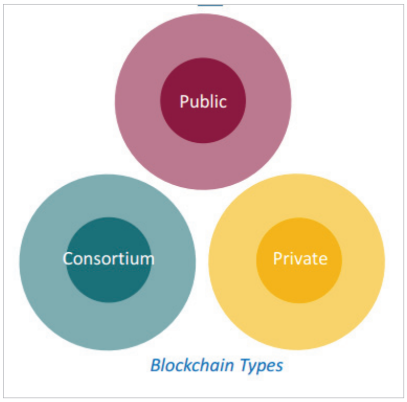

Figure 3: Types of blockchains

validate transactions on the network. Private blockchains are often used by businesses or organisations for internal purposes, such as supply chain management or record-keeping. They provide greater control and privacy compared to public blockchains but sacrifice some decentralisation.

Consortium blockchain: Consortium blockchains are a hybrid between public and private blockchains. They are managed and operated by a group of organisations or entities working together in a consortium. Consortium blockchains offer a more decentralised approach compared to private blockchains while maintaining a level of control among the consortium members. They are often used in industries where multiple stakeholders need to collaborate, such as banking consortia or supply chain networks.

Where is blockchain used?

Blockchain technology has a wide range of uses across various industries.

- ***Cryptocurrencies:*** Blockchain technology is most commonly associated with cryptocurrencies like Bitcoin and Ethereum. It enables secure and transparent peer-to-peer transactions without the need for intermediaries like banks.
- ***Supply chain management:*** Blockchain can be used to track and trace the movement of goods across the supply chain. It provides real-time visibility, enhances transparency, and ensures the integrity of data, reducing fraud, counterfeiting, and improving efficiency.
- ***Healthcare:*** Blockchain technology can securely store and share medical records, ensuring patient data privacy, interoperability, and accessibility. It can also streamline the process of clinical trials, drug traceability, and supply chain management in the healthcare industry.
- ***Finance and banking:*** Blockchain has the potential to revolutionise the financial sector by enabling faster, more secure, and cost-effective cross-border payments, remittances, and settlements. It can also facilitate identity verification, fraud prevention, and decentralised lending and borrowing platforms.
- ***Voting and governance:*** Blockchain can enhance the transparency, security, and integrity of voting systems. It enables verifiable and tamper-proof voting records, reducing the risk of fraud and ensuring fair and trustworthy elections.
- ***Intellectual property:*** Blockchain technology can be used to establish proof of ownership and protect intellectual property rights. It enables creators to securely register and manage their copyrights, patents, and trademarks, reducing infringement and facilitating licensing and royalty payments.

Pros and cons of blockchain

Even with all of its complexity, blockchain has virtually limitless potential as a decentralised method of record-keeping. Blockchain technology may very well have uses beyond those mentioned above, ranging from increased user privacy and heightened security to lower processing costs and fewer mistakes. There are a few downsides, though.

Pros

- ***Enhanced accuracy***: Blockchain eliminates the need for human involvement in verification processes, leading to improved accuracy.
- ***Cost reductions:*** By eliminating the necessity for third-party verification, blockchain reduces operational costs.
- ***Decentralisation:*** Its decentralised nature makes tampering with blockchain data more challenging.
- ***Secure and efficient transactions:*** Blockchain ensures secure, private, and efficient transactions.
- ***Financial inclusion:*** It serves as a banking alternative and provides a means to secure personal information, particularly beneficial in countries with unstable or underdeveloped governments.

Cons

- ***Technology costs:*** Some blockchains incur significant technology expenses.
- ***Low transaction throughput:*** Blockchains often exhibit low transactions per second.
- ***History of illicit use:*** Blockchain has been associated with illicit activities, including those on the dark web.
- ***Regulatory variations:*** Regulation of blockchain varies by jurisdiction and remains uncertain.
- ***Data storage constraints***: Blockchains may encounter limitations in data storage capacity. END

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